

Abivan Vijay

Robotics, embedded hardware and web applications

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PROFILE 14-year-old builder who designs and builds ESP32 hardware, autonomous robots and full-stack web applications, and takes them to national competitions, hackathons and expos. Uses AI coding tools (Claude Code, Codex) to take ideas from prototype to working build quickly.

ACHIEVEMENTS	National Robotics Championship 2026 , Delhi: drone soccer and innovation contest	Upcoming 20-21 Nov 2026
	Robotics for Good Youth Challenge (ITU): national round, Delhi	Upcoming 10 Oct 2026
	Robotics for Good Youth Challenge : selected for the national round, Senior category	Sep 2026
	PEC Hacks 4.0 : Consolation Prize, Vault (team lead)	Aug 2026
	School Shark Tank-style pitch competition : presented Volt, with all 14 branches competing	Aug 2026
	Young Environmental Scientist 2026 (IWMA with IIT Madras): participant, Sewer Gas Detector	Aug 2026
	National Robotics Championship : selected with CRASH, after qualifying at the SRM zonals	Jul 2026
	School-level expo : Winner, FreshSense	Feb 2026
	RoboWunder International Competition : participant at the Regionals, FreshSense	Jan 2026
	SRM University, Ramapuram expo : participant, Smart Guard	Nov 2025

PROJECTS	CRASH: Chennai Road Accident Safety Hub <i>Team project</i>	JavaScript, Python
	Web app that maps road accidents across Greater Chennai, ranks junctions by a severity-weighted risk score and recommends interventions, with analytics dashboards, Monte Carlo simulation and an AI data assistant.	
	Robotics for Good: Public-Health Response Robot <i>Team project</i>	LEGO SPIKE Prime
	Fully autonomous robot that runs 2-minute simulated emergency-response missions: moving samples, closing a quarantine zone, delivering medical kits and colour-sorting patients.	
	Vault: Cold-Chain Integrity Console <i>Team lead</i>	TypeScript, ESP32
	Monitors a vaccine shipment's cold chain and records every event to a hash-chained ledger that can be verified afterwards.	
	FreshSense: Food Spoilage Detection <i>Team lead, 3 members</i>	ESP32, Blynk IoT
	ESP32 conveyor system that classifies food as FRESH or ROTTEN using gas sensors and a self-calibrated baseline, with remote monitoring through Blynk IoT.	
	Also built: Sewer Gas Detector (ESP32 wearable for sanitation workers, ESP-NOW), Volt (peer-to-peer solar energy ledger, Open Energy Challenge 2026), Smart Guard (team lead, 3 members), WriteWise AI (Flask + Claude), Landmine Detection Shoe (prototype), RFID Door Lock.	

SKILLS	Microcontrollers and robotics: Arduino, ESP32, LEGO SPIKE Prime; servo control	
	Sensors and modules: gas, climate and RFID sensors; displays, servos	
	Wireless and IoT: Wi-Fi, ESP-NOW (no-router device links); Blynk IoT cloud dashboards	
	Languages: JavaScript, TypeScript, Python, Arduino C++, HTML, CSS	
	Backends and data: Node.js, Express, FastAPI, Flask; MongoDB, Supabase; REST APIs, Claude API, Resend	
	Web: analytics dashboards and maps; canvas and scroll-driven animation	
	Techniques: sensor calibration; hash-chained ledgers; Monte Carlo simulation	
	Tools: Claude Code, Codex; Git, GitHub; Render, Vercel	

EDUCATION	Velammal Academy, Nolambur, Chennai: Grade 9, CBSE (age 14)	2026-27
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Title Abivan Vijay, CV	Portfolio abivan-dev.vercel.app	Updated Oct 2026	Sheet 1 of 1
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